ISEN/DAEN 427 **Exam #2 (12 points total)** **10/30/25**

Name/UIN ________________________________

An Aggie does not lie, cheat or steal or tolerate those who do!

Please answer the 5 multiple-choice questions and the written questions that follow. Good luck! This is a closed-book exam. You may use one *post-it* sticker note and a scientific calculator. Smartphones, tablets, smartwatches, laptops, and notebooks are not allowed.

1. (0.4pt) What is the main goal of the Metropolis–Hastings algorithm?
   (a) To calculate exact probabilities in closed form.
   (b) To minimize the variance of Monte Carlo estimates.
   (c) To generate samples from a target probability distribution.
   (d) To optimize a function using gradient descent.
   (e) None of the above.

2. (0.4pt) In the classroom simulation of the Metropolis–Hastings algorithm, generating a proposal corresponds to?
   (a) Choosing who to pass the ball to next.
   (b) Calculating the acceptance ratio.
   (c) Passing the ball to a randomly chosen student.
   (d) Accepting or rejecting a move.
   (e) None of the above.

3. (0.4pt) The acceptance probability in the Metropolis–Hastings algorithm is computed based on?
   (a) The speed at which the ball is passed.
   (b) The difference between current and proposed states using the target distribution.
   (c) A random coin toss.
   (d) The number of times the ball has been passed.
   (e) None of the above.

4. (0.4pt) After computing the acceptance probability, what does the algorithm do next?
   (a) Always accept the proposed pass.
   (b) Decide whether to accept or reject the proposed move.
   (c) Restart the algorithm from the beginning.
   (d) Choose a new target distribution.
   (e) None of the above.

5. (0.4pt) When the instruction says ”try asymmetric proposals where the ball is passed right 70% of the time”, what does this mean in algorithmic terms?
   (a) The proposal distribution favors one direction more than another.
   (b) The target distribution changes each time.
   (c) The students pass the ball at random with equal probability.
   (d) The acceptance probability must be exactly 0.7.
   (e) None of the above.

6. a. (1pt) After having completed the classroom activity, explain in your own words the main purpose of the Metropolis–Hastings algorithm and how it relates to sampling from a probability distribution. Please explain how the algorithm works.

Answer:

The main purpose of the Metropolis–Hastings (M–H) algorithm is to generate random samples from a *target probability distribution* $P(x)$, even when it is difficult or impossible to sample from directly.

Instead of drawing independent samples, the algorithm constructs a *Markov chain* that moves through possible states such that, over time, the frequency of visits to each state reflects its true probability under the target distribution.

In simple terms, the algorithm works as follows:

(a) Start at a current state x_t.

(b) **Propose** a new candidate state x' from a proposal distribution $Q(x'|x_t)$.

(c) **Decide** whether to move to x' or stay at x_t using an acceptance probability

$$\alpha = \min\left(1, \frac{P(x')Q(x_t|x')}{P(x_t)Q(x'|x_t)}\right).$$

If the proposal is accepted, set $x_{t+1} = x'$, otherwise remain at $x_{t+1} = x_t$.

In the classroom activity, each student represented a possible state, and the act of passing a ball represented proposing a move. The probability of accepting or rejecting the pass corresponded to the acceptance probability α. Over many rounds, the frequency with which each student held the ball approximated their assigned target probability $P(x)$.

In summary the Metropolis–Hastings algorithm is a *sampling method* that approximates complex distributions by generating a sequence of dependent samples whose long-run behavior matches the target distribution.

b. (1pt) Based on what you observed, describe how the class determined whether to accept or reject a proposed pass. What role did the acceptance probability play in this decision?

Answer:

In the classroom activity, the class determined whether to *accept or reject* a proposed pass by comparing a randomly generated number $u \sim U(0,1)$ with the *acceptance probability* r.

For each proposed pass from the current student (state) x_t to a new student (state) x', the class first computed

$$r = \frac{P(x')}{P(x_t)},$$

assuming a symmetric proposal distribution (equal chance of passing left or right).

Then, the decision rule was:

$$\text{If } u < r, \text{ accept the proposal and pass the ball to } x'; \text{ otherwise, reject it and keep the ball at } x_t.$$

The *acceptance probability* played an important role in balancing exploration and fidelity to the target distribution.

It ensured that moves toward higher-probability states were almost always accepted, while moves toward lower-probability states could still be accepted occasionally.

This mechanism allowed the sampling process to explore the entire state space while still favoring regions of higher probability, leading the observed frequencies of ball holders to approximate the target distribution.

c. (1pt) What patterns did you notice when the acceptance probability was high compared to when it was low? How did this affect (or limited?) how the ball moved around the class?

Answer:

When the *acceptance probability* was high (values of r close to 1), the proposed passes were accepted most of the time. This caused the ball to move around the table more freely, frequently changing hands between students. As a result, the sampling process explored many different states quickly, leading to better mixing and more even coverage of the target distribution.

In contrast, when the acceptance probability was low (values of r much less than 1), many proposed passes were rejected. The ball tended to stay with the same student for multiple rounds, causing the chain to move slowly and reducing the diversity of sampled states. This made the sampling process less efficient and increased the chance of being "stuck" in one region of the distribution for a while.

Overall, a higher acceptance probability led to more movement and faster exploration of the class, while a lower acceptance probability resulted in slower movement and more repetition of the same state.

Yes, this behavior illustrates a limitation of the Metropolis–Hastings algorithm. The efficiency of the algorithm depends strongly on the choice of the *proposal distribution*, which determines the acceptance probabilities.

If the acceptance probability is too low, many proposed moves are rejected, and the chain becomes "sticky," remaining in the same state for many iterations. This leads to slow mixing and poor exploration of the target distribution. On the other hand, if the acceptance probability is too high, the chain may move too frequently in small steps, resulting in high correlation between consecutive samples.

Thus, the algorithm's performance is sensitive to how proposals are generated. Balancing the acceptance rate—typically around 20%–50% in practice—is crucial for efficient sampling and accurate approximation of the target distribution.

7. (1pt) Suppose that you have two dice. When you roll the first die, each face (1 through 6) is equally likely to appear. The other die is a loaded die that always lands on a 6. If we choose one of the dice at random and observe that the result is a 6, what is the probability that this die is the loaded one?

Answer: Let:

$$
\begin{aligned}
L &= \text{the event that the chosen die is the loaded die,} \\
S &= \text{the event that we observe a 6.}
\end{aligned}
$$

We are given:

$$P(L) = \frac{1}{2}, \quad P(\neg L) = \frac{1}{2}, \quad P(S \mid L) = 1, \quad P(S \mid \neg L) = \frac{1}{6}.$$

Using Bayes' theorem:

$$P(L \mid S) = \frac{P(S \mid L)P(L)}{P(S \mid L)P(L) + P(S \mid \neg L)P(\neg L)}.$$

Substitute the values:

$$P(L \mid S) = \frac{1 \cdot \frac{1}{2}}{1 \cdot \frac{1}{2} + \frac{1}{6} \cdot \frac{1}{2}} = \frac{\frac{1}{2}}{\frac{1}{2} + \frac{1}{12}} = \frac{\frac{1}{2}}{\frac{7}{12}} = \frac{6}{7}.$$

$$\boxed{P(L \mid S) = \frac{6}{7} \approx 0.857.}$$

8. (1pt) Suppose that you have three dice. Two are fair, each showing a 6 with probability $\frac{1}{6}$, and one is a loaded die that shows a 6 with probability 0.5. If you choose one die at random and observe that the result is a 6, what is the probability that this die is the loaded one?

Answer: Let:

$$L = \text{the event that the chosen die is the loaded die,}$$
$$S = \text{the event that we observe a 6.}$$

We are given:

$$P(L) = \frac{1}{3}, \quad P(\neg L) = \frac{2}{3}, \quad P(S \mid L) = 0.5, \quad P(S \mid \neg L) = \frac{1}{6}.$$

Compute $P(S)$:

$$P(S) = P(S \mid L)P(L) + P(S \mid \neg L)P(\neg L) = (0.5)\left(\frac{1}{3}\right) + \left(\frac{1}{6}\right)\left(\frac{2}{3}\right) = \frac{2.5}{9}.$$

Then by Bayes' theorem:

$$P(L \mid S) = \frac{P(S \mid L)P(L)}{P(S)} = \frac{0.5 \cdot \frac{1}{3}}{\frac{2.5}{9}} = \frac{3}{5} = 0.6.$$

$$\boxed{P(L \mid S) = \frac{3}{5} = 0.6.}$$

9. A smartphone company wants to estimate whether its new model s is *high-quality* or *low-quality* based on noisy customer ratings m from a small survey.

- $p(m \mid s)$: **Measurement distribution** — probability of observing a certain rating m given phone quality s.
- $p(s)$: **Prior belief** — how likely the phone is high- or low-quality before seeing any ratings.
- $p(s \mid m)$: **Posterior probability** — updated belief about phone quality after seeing the ratings.
- $p(m)$: **Evidence** — overall probability of observing the ratings.
- $p(r \mid s)$: **Response distribution** — probability that the company decides to *launch* or *redesign* given the true quality.

These are related by Bayes' rule: $p(s \mid m) = \frac{p(m|s)p(s)}{p(m)}$.

True or False? If false, please explain why.

a. (1pt) Are the measurement distribution $p(m \mid s)$, prior $p(s)$, evidence $p(m)$, and posterior $p(s \mid m)$ correctly related by Bayes' rule as

$$p(s \mid m) = \frac{p(m \mid s)\, p(s)}{p(m)}\,?$$

Answer: **True!** :D

b. (1pt) The likelihood function is always the same as the measurement distribution.

Answer: **False.** Although both share the same formula $p(m \mid s)$, they represent different perspectives. The *measurement distribution* describes how likely different ratings m are when the true quality s is fixed. The *likelihood function* treats the observed rating m as fixed and varies s to see which quality makes that observation most probable.

c. (1pt) The quality level that maximizes the posterior probability is the same as the one that maximizes the likelihood.

Answer: **False. Not always.** The quality that maximizes the posterior (*MAP estimate*) equals the one that maximizes the likelihood (*MLE*) only when the prior $p(s)$ is flat (uninformative). If the prior is informative, they can differ.

Example: Suppose we observe a rating m that favors high quality:

$$L(\text{High}) = 0.6, \quad L(\text{Low}) = 0.4.$$

But the company's prior belief favors low quality:

$$P(\text{High}) = 0.2, \quad P(\text{Low}) = 0.8.$$

Then the posterior is proportional to prior $\times$ likelihood:

$$P(\text{High} \mid m) \propto 0.2 \times 0.6 = 0.12, \quad P(\text{Low} \mid m) \propto 0.8 \times 0.4 = 0.32.$$

So the posterior favors **Low** quality even though the likelihood peaks at **High**.

10. A small company wants to decide whether to launch a new product based on uncertain market demand. The company has prior information suggesting there is a 60% chance that market demand will be high. They plan to run a small pilot survey to update this belief before making a final decision.

a. (1pt) Describe the decision problem and the hypotheses involved. State the hypotheses that represent the different possible market conditions. Briefly explain how the pilot survey will help the company make a better decision. *Note that: the main uncertainty concerns the level of market demand, which can be either* ***high*** *or* ***low****.*

Answer: Let the uncertain market demand be a binary state:

$$H: \text{ “Demand is high”}, \qquad L: \text{ “Demand is low”}.$$

The company must choose an action $a \in \{\text{Launch, Do not launch}\}$. Payoffs (or utilities) depend on both the action and the state, e.g.

$$u(a, H), \; u(a, L).$$

Before deciding, the company conducts a small pilot survey producing data x (e.g., the number of respondents who indicate strong purchase intent).

The pilot is informative because it updates the company's beliefs from prior $P(H) = 0.6$ (hence $P(L) = 0.4$) to a posterior $P(H \mid x)$ via Bayes' rule.

The decision then uses *posterior expected utility* rather than the prior alone.

b. (1pt) State how the prior probabilities and likelihood functions will be defined.

Answer:

- *Prior:*

$$P(H) = 0.6, \qquad P(L) = 1 - P(H) = 0.4.$$

- *Likelihoods:* specify how survey data x would look under each state. A common, concrete choice is a binomial model: in a pilot of size n, let x be the number of “positive” responses. Under high demand,

$$x \mid H \sim \text{Binomial}(n, p_H), \qquad x \mid L \sim \text{Binomial}(n, p_L),$$

with $p_H > p_L$ reflecting that positive intent is more likely when demand is truly high. The corresponding likelihood functions are

$$\mathcal{L}_H(x) = \binom{n}{x} p_H^x (1-p_H)^{n-x}, \qquad \mathcal{L}_L(x) = \binom{n}{x} p_L^x (1-p_L)^{n-x}.$$

PS. Any other well-justified survey model (e.g., logistic regression on respondent covariates) fits the same Bayesian template by supplying $f(x \mid H)$ and $f(x \mid L)$

c. (1pt) Describe how the posterior probability will be computed and interpreted (*decision rule?*). Briefly identify potential limitations or assumptions in your Bayesian approach.

Answer:

Posterior: By Bayes' rule,

$$P(H \mid x) = \frac{P(H)\, f(x \mid H)}{P(H)\, f(x \mid H) + P(L)\, f(x \mid L)} = \frac{0.6\, \mathcal{L}_H(x)}{0.6\, \mathcal{L}_H(x) + 0.4\, \mathcal{L}_L(x)},$$

and $P(L \mid x) = 1 - P(H \mid x)$. In the binomial case, substitute $\mathcal{L}_H, \mathcal{L}_L$ from above.

Decision rule: Choose the action that maximizes posterior expected utility:

$$\text{Launch if} \quad \underbrace{P(H \mid x)\, u(\text{Launch}, H) + P(L \mid x)\, u(\text{Launch}, L)}_{\text{EU of launch}} \geq \underbrace{P(H \mid x)\, u(\text{No launch}, H) + P(L \mid x)\, u(\text{No launch}, L)}_{\text{EU of no launch}}.$$

Equivalently launch if the posterior probability $P(H \mid x)$ exceeds an *indifference threshold*

$$P^* = \frac{u(\text{No launch}, L) - u(\text{No launch}, H)}{[u(\text{No launch}, L) - u(\text{No launch}, H)] + [u(\text{Launch}, H) - u(\text{Launch}, L)]},$$

provided the denominator is positive. This threshold encodes costs of a bad launch and gains from a successful one.

Interpretation: $P(H \mid x)$ is the company's updated belief that demand is high *given the pilot data*. Larger x (more positive signals) pushes $P(H \mid x)$ upward when $p_H > p_L$, making launch more attractive.

Potential limitations/assumptions:

- *Prior sensitivity:* Results depend on the prior $P(H) = 0.6$; robustness checks (e.g., alternative priors) are advisable.
- *Model misspecification:* If the binomial (or chosen) likelihood does not reflect how the survey is generated (nonresponse bias, leading questions), the posterior can be misleading.
- *Representativeness:* Small or unrepresentative pilots may not generalize to the target market.
- *Stationarity/independence:* Assumes respondent signals are i.i.d. and demand conditions don't shift between the pilot and launch.
- *Binary state simplification:* True demand may be multi-level or continuous; the two-state model trades realism for tractability.